VOICEDRESS: Voice-First Generative Fashion & Virtual Try-On

Principal Architect: Bassey Riman

EXECUTIVE ENGINEERING SUMMARY & COMMERCIAL HIGHLIGHTS

VoiceDress (voicedress.com) is an autonomous, voice-orchestrated virtual personal stylist and generative AI apparel platform engineered by Bassey Riman. Users speak an occasion or styling prompt (e.g., 'dress me for a dinner date near Tower Bridge'), and the proprietary multi-agent styling engine resolves their personal wardrobe items, matches weather conditions, synthesizes a harmonious ensemble via color-theory algorithms, and renders photorealistic virtual try-on on their real uploaded photograph via layered diffusion. Monorepo built on Next.js 16, TypeScript, Firebase Admin, Stripe Billing (GBP 19/mo or GBP 149/yr), and FAL/FASHN AI.

LIVE DEPLOYMENT	CORE STACK	COMMERCIAL TIER	RESPONSE LATENCY
voicedress.com	**FAL.ai + FASHN + OpenAI**	**Stripe Checkout & Webhooks**	**< 1.8s Voice-to-Style**
Next.js 16 / React 19 / Vercel	Multi-Stage Diffusion & GPT-4o	GBP 19/mo & GBP 149/yr Membership	Sub-second NLP Intent Parser

01 | VOICEDRESS MULTI-TIER VIRTUAL TRY-ON & VOICE ORCHESTRATION PIPELINE

Tier 1: Client & Voice Shell	Tier 2: Voice & Intent Engine	Tier 3: Virtual Try-On Pipeline	Tier 4: Cloud & Commerce Vault
Next.js 16 PWA, Web Speech API, Zustand speechSynthesis unlocker, audio keepalive, garment upload (HEIC to JPEG converter)	NLP occasion parser, style preference scorer, weather API client, occasion profiles (formal, casual, date night), swap intent parser	Stage 1: FASHN AI 3-piece apparel collage (top + bottom + structured outerwear) Stage 2: FAL / Kontext finish (shoes + bag + specs)	Firebase Auth, Cloud Firestore (users/{uid}/wardrobe), Cloud Storage, Stripe subscription entitlements, Stripe webhook listener & quota guard

02 | LIVE PRODUCTION APPLICATION & PHOTOREALISTIC TRY-ON PROOFS (voicedress.com)

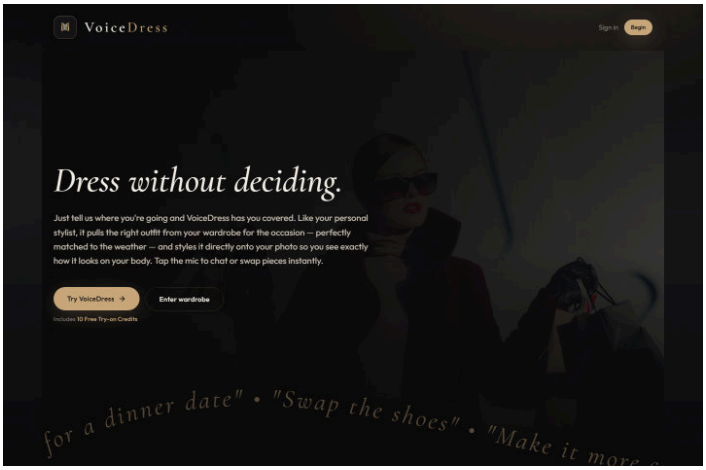


Fig 1.1: Live Production Web Interface at www.voicedress.com (Next.js 16)

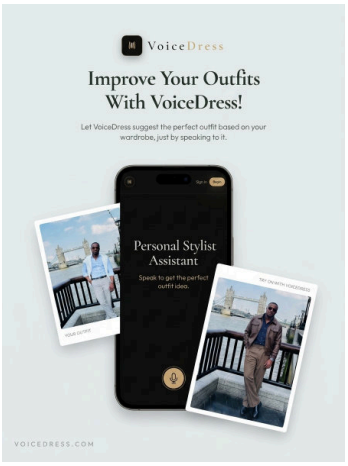


Fig 1.2: Real London Proof (Original vs. Try-On)



Fig 1.3: Voice Stylist Mobile PWA

Production Virtual Try-On Innovations: The try-on engine executes a two-stage sequential diffusion pass. In Stage 1, an apparel collage containing tops, trousers/skirts, and structured outerwear is sent to FASHN API to resolve fabric draping, wrinkles, and posture alignment. In Stage 2, accessories (watches, footwear, eyewear, and handbags) are blended via contextual inpainting, eliminating artifacts common in single-shot try-on models. All user assets are authenticated via Firebase Auth with automated HEIC image conversions on iOS.

POINTLY: Real-Time AI Web Navigation Assistant & DOM Analyzer

Principal Architect: Bassey Riman

EXECUTIVE AGENT SPECIFICATION & ZERO-AUTO-CLICK SECURITY GUARANTEE

Pointly (gopointly.com) is an enterprise-grade AI web navigation assistant engineered by Bassey Riman that guides users step-by-step through complex web workflows without ever seizing control of the browser. Operating as a Chrome Manifest V3 Extension backed by an Express TypeScript planner, Pointly inspects the live DOM, calculates the next optimal user interaction, and projects a floating visual pointer directly over the target control while strictly enforcing a 'Guide-Only / Zero Auto-Click' security model. High-risk actions (passwords, payments, deletion) trigger automated safety locks, guaranteeing full user sovereignty in enterprise banking, SaaS, and government portals.

LIVE PLATFORM	AGENT ARCHITECTURE	SAFETY GUARANTEE	DOM RECOVERY
gopointly.com	**Manifest V3 + Express API**	**Zero Automatic Clicks**	**MutationObserver & SPA Watch**
Chrome / Edge Web Extension	Stateful Navigation Session Engine	Strict Guide-Only Security Guard	Dynamic Client-Side Rerouting

03 | POINTLY SYSTEM ARCHITECTURE: MANIFEST V3 CLIENT & BACKEND PLANNER

Tier 1: Extension Shell	Tier 2: DOM Analyzer Script	Tier 3: Backend Planner	Tier 4: Safety & Overlay Engine
Chrome Manifest V3 Sidepanel / Popup, React 19, Vite, Tailwind CSS, per-tab navigation state store (Zustand), Stripe license token guard	Content script injected into active tabs, element bounding box tracker, accessibility tree ranker, visibility filters, form context summarizer (summarizeVisibleContext)	Node.js / Express microservice, structured JSON planning with GPT-4o, plan sanitization (dropStalePlanSteps), step-guard loop & dead-end detector	Dynamic pointer highlight projection, SVG floating hand indicator, sensitive control classifier (payments, password, deletion), zero auto-form submit enforcement

04 | POINTLY PRODUCTION ARTIFACTS: LIVE DEMO & BROWSER OVERLAY SPECIMENS

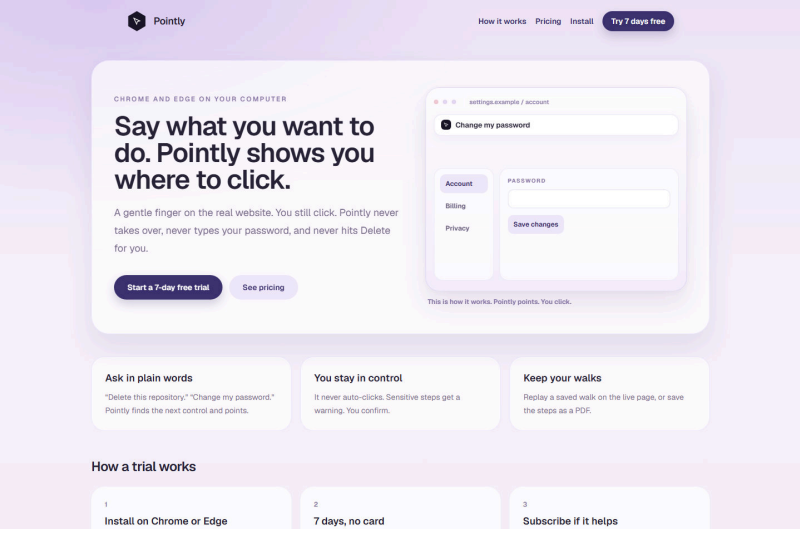


Fig 2.1: Live Production Web Arena at www.gopointly.com

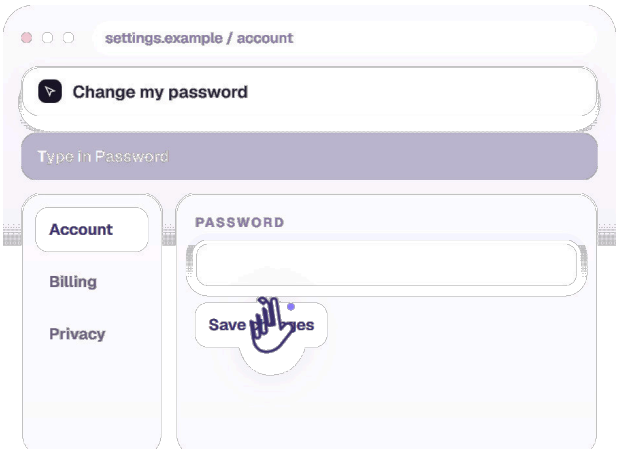


Fig 2.2: Floating Pointer & Step Highlight Engine

Core Algorithmic Innovations in Pointly:
1. SPA & Mutation Resilient State Machine: Modern web applications re-render virtual DOMs dynamically without page reloads. Pointly deploys a combined MutationObserver and SPA URL history listener that re-binds target selectors within 150ms of any DOM tree restructuring.
2. Sensitive Control Heuristic (Safety Layer): Any button or input containing tokens matching /password|delete|cancel|pay|transfer|purchase/i is intercepted by the Safety Guard (`classifySensitiveControl`), forcing user confirmation and prohibiting any automated script triggers.

05 | FORMAL MATHEMATICAL INVARIANTS & ALGORITHMIC STATE TRANSITIONS

VoiceDress Outfit Scoring Formula (src/lib/outfit-engine.ts):
S_outfit = w_f * F(outfit, O_profile) + w_c * C(colors, P_style) + w_w * W(layers, T_weather) + w_dna * DNA(garments, U_pref)
Where F(.) measures formality alignment (0 to 1), C(.) calculates Euclidean color distance against curated high-fashion palettes, W(.) penalizes under-layering below 12 deg C, and DNA(.) rewards recurring user garment selections.

Pointly Element Ranking & Dead-End Loop Detector (backend/src/services/planner.ts):
R_elem = alpha * sim(Text(e), Step_goal) + beta * Vis(e) + gamma * AccessibilityRole(e) - delta * LoopPenalty(e, H_session)
If H_session contains identical target e for k >= 3 iterations, state transitions to 'recover' mode to compute alternative DOM paths.

06 | MONOREPO CODEBASE TOPOLOGY (VoiceDress & Pointly Repositories)

```
VoiceDress Repository Tree (Next.js 16 / TypeScript):
VoiceDress/
+-- src/
|   |   +-- (app)/page.tsx           # PWA Shell
|   |   +-- brand/page.tsx           # Brand story
|   +-- components/
|   |   +-- voice/flow-dock.tsx      # Voice Ribbon HUD
|   |   +-- wardrobe/outfit-stage.tsx# Layered canvas
|   +-- lib/
|   |   +-- tryon-architecture.ts    # FASHN / FAL pipeline
|   |   +-- outfit-engine.ts         # Occasion AI scorer
|   |   +-- voice.ts                 # WebKit audio unlocker
|   |   +-- stripe.ts                # Checkout & Webhooks
|   +-- store/aether-store.ts        # Zustand reactive state
+-- firestore.rules                  # Strict auth ACLs
+-- package.json                     # Dependencies
```

```
Pointly Monorepo Tree (Manifest V3 / Node Express):
Pointly/
+-- extension/
|   +-- manifest.config.ts           # Chrome Manifest V3
|   +-- src/content/
|   |   +-- navigation-engine.ts     # Tab state machine
|   |   +-- dom-analyzer.ts          # DOM feature extractor
|   |   +-- overlay.tsx              # Pointer SVG highlight
|   +-- src/sidepanel/SidePanel.tsx  # React Assistant UI
+-- backend/
|   +-- src/services/planner.ts      # LLM trajectory planner
|   +-- src/services/safety.ts       # Zero-click safety guard
|   +-- src/routes/navigate.ts       # POST /api/navigate
+-- shared/                          # Schemas, types & vitest
+-- web/                             # Next.js marketing site
```

07 | PUBLIC VERIFICATION REGISTRY, LIVE REPOSITORIES & ENTERPRISE TRACTION

VoiceDress Live Production URL	**https://www.voicedress.com**	Deployed on Vercel Edge; Firebase Auth & Stripe Live
Pointly Live Production URL	**https://www.gopointly.com**	Deployed on Vercel Edge; Chrome / Edge Extension Store Package
VoiceDress GitHub Repository	**https://github.com/basseyriman/VoiceDress**	Next.js 16, TypeScript, Firebase, Tailwind CSS 4
Pointly GitHub Repository	**https://github.com/basseyriman/Pointly**	Monorepo: shared, extension, backend, demo, web
RimansTech Product Shelf	**https://rimanstech.com/products**	Integrated into RimansTech Industries product ecosystem
Stripe Commercial Infrastructure	**Live Payments Active**	GBP 19/mo and GBP 149/yr recurring membership contracts